

KIRCHHOFF’S LAW OF THERMAL EMISSION WITH BUILT-IN UV/IR CUTOFFS FROM HQIV’S DISCRETE NULL LATTICE

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ABSTRACT. Position in the publication chain. This is the third paper in the Tier-1 foundation-extension trilogy: the HQIV main paper [2] fixes the framework, the lightcone paper [3] introduces the discrete null-lattice readout, the closure paper [5] certifies the $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ Lie closure, the 3D causal-growth paper [6] isolates the dimensional hinge and the rational (α_d, γ_d) family, and the octonionic-action paper [4] packages the variational layer that we ride here for the $\alpha = 3/5$ imprint and the patch-ontology reader contract. The Tier-0 records and the Tier-1 companions [6, 4] carry minted Zenodo DOIs; this manuscript and the remaining Tier-1 queue (`thermodynamics.arrow/`) co-deposit as the Lean PR builds clear.

Result. Kirchhoff’s law of thermal emission is recovered from two HQIV axioms—a discrete null lattice with Planck cell size and informational monogamy ($\alpha = 3/5$)—with a *finite* mode count at every shell, given in closed form by the stars-and-bars identity

$$\mathcal{N}(m) = (m+2)(m+1), \quad \sum_{m=0}^n \mathcal{N}(m) = \frac{1}{3}(n+1)(n+2)(n+3).$$

The resulting Planck spectrum is a *finite sum* between an ultraviolet cutoff at $m = 0$ (the Planck pole, $\omega_{\text{PI}} = T_{\text{PI}}$) and an arbitrary infrared cutoff at $m = M$ (the lock-in reference shell). No ultraviolet catastrophe arises, no renormalisation is performed, and no external blackbody calibration is imposed: the mode budget is *finite by construction*, not by regularisation. This is fully consistent with the patch-theory reader contract imported below [4]: the discrete patch layer is the ontology, and what the literature calls “the continuum limit” is admitted only as a calculation-approximation rewrite for cross-citation, never as a foundation.

Sourcing of the dimensionful inputs. The proton mass at the lock-in shell $m = 4$ uses the theory’s internal constituent-plus-binding readout, with the observed value $M_p = 938.272 \text{ MeV}$ serving as the sole dimensionful anchor for the mass chart (the [proton lock-in](#) convention, recorded in the [scale-witness](#) formalism of [7]). The wall-clock age $t_{\text{wall}} = 51.2 \text{ Gyr}$ is an *output* of the HQIV-modified CLASS implementation—a discrete null-lattice background with the ADM lapse $N(t) = 1 + \Phi + \varphi t$ and φ -dependent evolution (the [HQVM wall-clock compression](#))—not a free parameter. The only externally measured dimensionful input used downstream of these is FIRAS $T_{\text{CMB}} = 2.7255 \text{ K}$ [8].

One quantitative prediction under the chosen witness. With the proton lock-in scale-witness fixing the mass/unit chart, all other dimensionful quantities—CODATA $1/\alpha$, CMB horizons, cosmic birefringence—become *predictions or comparison layers*. We exhibit one such prediction here:

- **Isotropic cosmic birefringence at present epoch.** The birefringence calculation uses the HQIV wall-clock age above together with the lattice mode count $\mathcal{N}(m)$ under the propagation-shell identification (§7), giving $\beta_{\text{HQIV}} = 0.379^\circ$. This agrees with Planck PR4 EB ($\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ [1]) within -0.40σ . Substituting HQIV’s apparent age 13.8 Gyr for the wall-clock age gives 0.103° (-2.55σ): the prediction *discriminates* HQIV’s two internal time outputs.

Status. The derivation chain is mechanised sorry-free against mathlib in the public slimmed Lean mirror [7]; reader-friendly anchors with clickable Lean links live in Appendix A. The Friedmann radiation-era scaling $H \propto T^2$ is also recovered, not imported: it follows from the quadratic asymptotic form of $\mathcal{N}(m)$ under the propagation-shell identification. All numerical results in this paper (the $\beta = 0.379^\circ$ prediction, the wall-clock-vs-apparent-age discriminator, and the finite-mode Kirchhoff energy density at FIRAS T_{CMB}) are reproduced by the supplementary script bundle described in Appendix B, packaged as `scripts.zip` for co-deposit with the manuscript on Zenodo.

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. SCOPE AND CLAIM DISCIPLINE

This paper presents a derivation of Kirchhoff’s law of thermal emission inside HQIV. We do *not* reprove the classical quantum statistical mechanics; we replace its mode-density input with HQIV’s finite stars-and-bars count and show that the resulting blackbody machinery (i) is finite at every step, (ii) admits explicit UV and IR cutoffs by construction, and (iii) supports one downstream quantitative prediction (β at present epoch) that survives the data, under HQIV’s own scale-witness discipline.

- (1) **Two axioms.** Discrete null lattice with Planck cell size; informational monogamy fixing $\alpha = 3/5$ via the hockey-stick identity. These two are the only HQIV combinatorial inputs. The closed Lie algebra spanned by $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ used downstream is imported as a black-box

certificate from the closure paper [5]; the rational $(\alpha, \gamma) = (3/5, 2/5)$ pair used here is forced by the same monogamy ledger that the 3D causal-growth paper [6] derives combinatorially.

- (2) **Finite by construction.** All spectral sums are $\sum_{m \in [m_{\text{UV}}, m_{\text{IR}}]}$ over finite shell intervals in $\mathbb{N}$. No limits, no analytic continuation, no cutoff-removal procedure. In particular, no programme of taking a continuum mesh-refinement limit is invoked or required: per the patch-theory reader contract above, several load-bearing results would be erased rather than refined by such a limit (the rational α , the integer cycle index, the discrete ladder that the predictive content runs on).
- (3) **One dimensionful anchor.** The HQIV formalisation [7] allows only one dimensionful calibration per phenomenology pass; we take the default [proton lock-in](#). At shell $m = 4$ the proton mass follows the internal constituent-plus-binding readout below, calibrated to the measured value $M_p = 938.272 \text{ MeV}$ to fix the mass/unit chart. The shell index $m = 4$ is the QCD-onset identification (not a free parameter), and the binding closure at that shell is a non-trivial HQIV calculation, but the MeV scale itself is an *anchor*, not an independent two-axiom output. Section 6 states what is and is not claimed.
- (4) **Wall-clock age from the HQIV-modified CLASS implementation.** The age $t_{\text{wall}} = 51.2 \text{ Gyr}$ used in the β prediction is an output of the HQIV-modified CLASS implementation (a discrete null-lattice background with the ADM lapse $N(t) = 1 + \Phi + \varphi t$ and φ -dependent evolution; the [HQVM wall-clock compression](#)), not a free parameter. Appendix A carries the Lean snapshot pointer.
- (5) **Predictions vs anchor.** Under the proton lock-in scale-witness, observables such as CODATA $1/\alpha$, cosmic birefringence β , and CMB horizon scales are *predictions or comparison layers*, not simultaneous anchors. This paper exhibits β as one such prediction.
- (6) **β is non-adjustable under the chosen witness.** Given the anchor, β_{HQIV} uses only the HQIV stars-and-bars count, the HQIV wall-clock age from the HQIV-modified CLASS implementation, and FIRAS T_{CMB} [8]. No birefringence-side anchor and no Friedmann equation enter the calculation. The internal discriminator wall-clock vs apparent age is a single-shot test.

2. BACKGROUND: CLASSICAL KIRCHHOFF AND THE NORMALISATION PROBLEM

Kirchhoff’s law of thermal emission [9] states that, at thermodynamic equilibrium, the spectral radiance emitted per mode equals the spectral radiance absorbed per mode—the universal blackbody spectrum. Two ingredients enter:

- (1) Detailed balance per mode (microscopic reversibility);
- (2) A mode-density per frequency interval $dN/d\nu$.

In the standard derivation, (i) is statistical-mechanical and robust; (ii) is the problematic input. Rayleigh–Jeans gives $dN/d\nu \propto \nu^2$ in a 3D cavity, unbounded above, and so the classical mean-energy multiplication $k_{\text{B}}T \cdot dN/d\nu$ diverges as $\nu \rightarrow \infty$ —the ultraviolet catastrophe. Planck’s quantum cure [11] provides $\langle E \rangle = h\nu / (e^{h\nu/k_{\text{B}}T} - 1)$, which exponentially suppresses high- ν contributions and yields a finite total energy.

This works numerically. But the cure is asymmetric: *the high-frequency cutoff is contributed by occupation suppression, not by mode counting*. In particular, nothing about the classical derivation prevents the existence of modes at arbitrarily high frequency; one simply trusts the Bose factor to render them harmless. This is the source of the normalisation question — if the mode count itself is unbounded, any quantum-gravity or trans-Planckian completion has to act on the mode list, not just on the occupation.

HQIV makes the mode count finite at the start.

3. HQIV MODE-COUNT AXIOM

Definition 3.1 (Discrete null-lattice mode count). *Let $\mathcal{N}(m)$ be the count of new Planck-scale lattice cells at temperature-ladder depth $m \in \mathbb{N}$. HQIV's discrete null-lattice axiom gives*

$$(1) \quad \mathcal{N}(m) = (m+2)(m+1).$$

This is the integer-solution count for $x+y+z=m$, $x, y, z \geq 0$, twice the binomial $\binom{m+2}{2}$.

Proposition 3.2 (Hockey-stick cumulative count).

$$(2) \quad \sum_{m'=0}^n \mathcal{N}(m') = \frac{1}{3}(n+1)(n+2)(n+3).$$

Proof. By induction on n , using (1); the [hockey-stick cumulative-count identity](#) states the equivalent integer form $3 \sum_{m'=0}^n \mathcal{N}(m') = (n+1)(n+2)(n+3)$. $\square$

Corollary 3.3 (Both cutoffs finite). *For any $0 \leq m_{\text{UV}} \leq m_{\text{IR}} < \infty$,*

$$\sum_{m \in [m_{\text{UV}}, m_{\text{IR}}]} \mathcal{N}(m) < \infty.$$

This is the structural fact that lets the entire spectral machinery be a finite sum, regardless of the temperature and without any regularisation procedure.

4. HQIV TEMPERATURE AND FREQUENCY LADDERS

The HQIV harmonic ladder fixes the shell-indexed frequency at:

$$(3) \quad \omega_m = \frac{T_{\text{Pl}}}{m+1}, \quad m \in \mathbb{N},$$

with $\omega_0 = T_{\text{Pl}}$ at the Planck pole (highest frequency, no UV catastrophe) and $\omega_m \rightarrow 0$ as $m \rightarrow \infty$ (IR limit, only $\mathcal{N}(m) \sim m^2$ modes each). Equivalently, the temperature ladder is $T(m) = T_{\text{Pl}}/(m+1)$, exposed as the [auxiliary-field temperature ladder](#).

5. KIRCHHOFF'S LAW ON A FINITE SHELL LIST

We now state Kirchhoff's law in HQIV form. Two ingredients, both finite by construction.

Definition 5.1 (Per-shell Bose occupation). *At temperature $T > 0$ and shell $m \in \mathbb{N}$,*

$$(4) \quad n_B(\omega_m, T) = \frac{1}{\exp(\omega_m/T) - 1}.$$

This is the standard Bose factor on one HQIV mode—the [per-shell Bose factor](#). It is positive and finite at every $m \in \mathbb{N}$ and $T > 0$ (with the divergence at $\omega_m = T \rightarrow 0$ outside the shell-discrete sector).

Definition 5.2 (Per-shell spectral energy).

$$(5) \quad E_m(T) = \mathcal{N}(m) \cdot \omega_m \cdot n_B(\omega_m, T).$$

Formalised as the [per-shell spectral-energy density](#). Every factor is positive at every m, T ; no regulariser is applied.

Definition 5.3 (Truncated blackbody energy density). *For any $m_{\text{UV}} \leq m_{\text{IR}}$ in $\mathbb{N}$,*

$$(6) \quad u(T; m_{\text{UV}}, m_{\text{IR}}) = \sum_{m=m_{\text{UV}}}^{m_{\text{IR}}} E_m(T).$$

Formalised as the [truncated blackbody energy density](#).

Theorem 5.4 (Finite-mode Kirchhoff law). *For any $T > 0$ and any $m_{\text{UV}} \leq m_{\text{IR}}$ in $\mathbb{N}$, (6) is a finite, positive, monotone sum:*

- (1) $0 \leq u(T; m_{\text{UV}}, m_{\text{IR}}) < \infty$;
- (2) $0 < u(T; m_{\text{UV}}, m_{\text{IR}})$ whenever $m_{\text{UV}} \leq m_{\text{IR}}$;
- (3) monotone in both cutoffs: lowering m_{UV} or raising m_{IR} adds non-negative contributions.

The result is true without any limiting procedure or renormalisation: it holds as a closed-form identity on $\mathbb{N}^2 \times \mathbb{R}_{>0}$.

Proof. The [truncated blackbody energy density](#) is non-negative and finite as a finite-range sum of non-negative terms; it is strictly positive on every nonempty shell window and monotone in both cutoff arguments. All three parts are zero-sorry Lean theorems in the same module (see [Appendix A](#)). $\square$

Remark 5.5 (The UV catastrophe never appears). *Unlike the classical derivation, no high-frequency divergence exists to suppress. $\mathcal{N}(m)$ is exactly $(m+2)(m+1)$ at the Planck pole $m=0$, namely 2 modes (lattice points $(0,0,1)$, $(0,1,0)$, $(1,0,0)$ scaled to two orientations), and there is no shell above it: $\omega > T_{\text{Pl}}$ does not exist in the HQIV mode list. The Bose factor in (4) additionally provides exponential suppression at fixed $T \ll T_{\text{Pl}}$, so the Planck spectrum shape is preserved everywhere classical spectroscopy probes; but the structural finiteness is independent of any thermal cutoff.*

Remark 5.6 (The IR is also finite). *At fixed temperature T , the Bose factor diverges as $\omega_m \rightarrow 0$ ($m \rightarrow \infty$): $n_B(\omega_m, T) \sim T/\omega_m = T(m+1)/T_{\text{Pl}}$, while $\omega_m \rightarrow 0$. The product $\omega_m n_B(\omega_m, T) \rightarrow T$, finite per mode; multiplying by $\mathcal{N}(m) \sim m^2$ gives a per-shell contribution growing as $m^2 T$. If one took $m_{\text{IR}} \rightarrow \infty$ the sum would diverge, exactly the classical Rayleigh–Jeans IR behavior—but HQIV’s discrete null lattice places a structural cutoff at the lock-in reference shell M (the [lock-in IR cutoff](#)), making the IR also finite at the lattice level, not just by stipulation.*

Definition 5.7 (Vacuum zero-point cell).

$$(7) \quad u_0(m_{\text{UV}}, m_{\text{IR}}) = \sum_{m=m_{\text{UV}}}^{m_{\text{IR}}} \frac{\mathcal{N}(m) \omega_m}{2}.$$

The [vacuum zero-point cell](#) is non-negative on every shell interval, by the same finite-sum-of-non-negatives argument used in Theorem 5.4.

Theorem 5.8 (Decomposition).

$$(8) \quad u_{\text{tot}}(T; m_{\text{UV}}, m_{\text{IR}}) = u_0(m_{\text{UV}}, m_{\text{IR}}) + u(T; m_{\text{UV}}, m_{\text{IR}}).$$

Formalised as the [blackbody-plus-vacuum energy density](#), with the additive decomposition lemma in the same module. No renormalisation is required to handle the zero-point cell—it is a finite, structural part of the spectrum on a finite shell list.

Summary. Equation (8) is the HQIV statement of Kirchhoff’s law. The standard Stefan–Boltzmann scaling $u(T; 0, M) \sim T^4$ and the Wien displacement bound emerge from (6) via the [horizon Stefan–Boltzmann module](#) and the [horizon Wien-displacement module](#), and rely only on Theorem 5.4. No regularisation appears in any step.

6. THE SCALE ANCHOR: PROTON MASS AT THE LOCK-IN SHELL $m = 4$

We use this section to be explicit about what the proton mass *is* and *is not* doing in this paper, since careless phrasing here can over-claim independence.

Proton lock-in as the mass chart. The HQIV library [7] is written so that a given phenomenology pass fixes *one* dimensionful scale before derived readouts are compared to experiment. We use the default, [proton lock-in](#): at shell $m = 4$ the proton mass comes from the internal constituent-plus-binding readout in the next paragraph, and the measured value 938.272 MeV is the calibration that sets the mass/unit chart for this paper—not a parameter we adjust elsewhere in the derivation. Once that chart is fixed, CODATA $1/\alpha$, CMB horizon scales, and PDG centrals [10] enter only as predictions or comparison checks. The [scale-witness](#) enum in Lean makes the same rule explicit: proton mass, fine-structure data, and a separate cosmological scale must not be treated as simultaneous inputs unless the active anchor is stated.

Shell index $m = 4$: chosen, not derived. The reference shell $m = 4$ is the QCD-onset identification: it is the HQIV temperature-ladder index where colour confinement and the lock-in chart begin. The choice is not a free fit parameter—the hadronic mass content of the universe requires *some* QCD-onset shell, and $m = 4$ is the smallest integer consistent with the small number of post-QCD steps needed to reach the nucleon harmonic. But it is also not a two-axiom output: the two axioms by themselves do not single out $m = 4$.

What *is* non-trivial at $m = 4$. Given the shell choice, the readout

$$(9) \quad M_p^{\text{HQIV}} = M_p^{\text{const}} - E_{\text{bind}}^{\text{shared}} = 938.272 \text{ MeV}$$

packages the nucleon as constituent-energy minus shared binding, formalised as the [derived proton mass](#) (with the [shared-harmonics closure](#) establishing (9)). The constituent harmonics M_p^{const} are dictated by the mode count (1) at $m = 4$; the shared-binding energy $E_{\text{bind}}^{\text{shared}}$ has its own composite-trace construction. These combinatorial structures are genuine HQIV content, and the closure to a single number 938.272 MeV is a non-trivial constraint on the lock-in shell-set. The [proton-anchor mass constant](#) records this as the active scale witness.

Why this is not a “test” in the same sense as β . At the level of the broader theory, the numerical value of M_p^{HQIV} is the chart-pinning anchor: it defines the unit of mass/energy used to interpret every other dimensionful quantity. Calling it a “test” would conflate two distinct roles: (i) the narrow constituent-plus-binding readout at the lock-in shell, which is a genuine HQIV calculation, vs (ii) the position of the lock-in shell on the integer ladder, which is a modelling choice in the broader theory. The honest framing is: M_p is the *anchor*, established once; β is the *prediction*, evaluated against the anchor and against the data.

Cross-check status. Within HQIV’s own discipline, candidate scale-witness alternatives are recorded (CODATA α witness, present-CMB witness) and exist in the [scale-witness enum](#). Switching to a different witness is a deliberate operation and would re-pin every dimensionful HQIV output. For this paper all numerics are reported under the proton lock-in scale-witness.

7. QUANTITATIVE TEST: COSMIC BIREFRINGENCE $\beta = 0.379^\circ$

This is the section where HQIV makes a single-shot quantitative prediction under the chosen scale witness. The proton-mass anchor of §6 has already pinned the unit chart. The birefringence calculation then combines the HQIV wall-clock age $t_{\text{wall}} = 51.2 \text{ Gyr}$ —an output of the HQIV-modified CLASS implementation (a discrete null-lattice background with the ADM lapse $N(t) = 1 + \Phi + \varphi t$ and φ -dependent evolution; the [HQVM wall-clock compression](#))—with the same stars-and-bars mode count $\mathcal{N}(m) = (m+2)(m+1)$ of (1) under the propagation-shell identification. Only one externally measured dimensionful input enters at this stage: FIRAS $T_{\text{CMB}} = 2.7255 \text{ K}$ [8]. No birefringence-side anchor enters—in particular, no Planck PR4 calibration [12]—and no Friedmann equation is imposed.

The Planck DR4 EB analysis of Eskilt & Komatsu [1] reports an isotropic cosmic birefringence

$$(10) \quad \beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ \quad (\text{Planck PR4 NPIPE, EB cross-spectrum}),$$

a $\sim 3.6\sigma$ detection, frequency-independent across the 30–353 GHz Planck sub-bands, isotropic within errors.

Two shell coordinates. HQIV carries two genuinely distinct shell coordinates (§3, §4, and the [cosmological shell-ladder module](#)):

$$(11) \quad m_T(T_{\text{obs}}) = T_{\text{Pl}}/T_{\text{obs}} - 1 \quad (\text{real-valued shell index in the temperature-ladder chart}),$$

$$(12) \quad m_{\text{prop}} = \text{count of HQIV propagation cells crossed since the Planck pole.}$$

A photon’s proper time is zero, so it does not accumulate temperature-ladder positions along its worldline; the cumulative birefringence formula uses the propagation coordinate.

Propagation-shell coarseness. Identifying one HQIV propagation shell at observation depth m_T with the bundle of new lattice simplices appearing at that depth,

$$(13) \quad \text{Planck cells per propagation shell} = \mathcal{N}(m_T),$$

the propagation-shell offset at the observation event is

$$(14) \quad m_{\text{prop}} = \frac{t_{\text{wall}}/t_{\text{Pl}}}{\mathcal{N}(m_T)} = \frac{t_{\text{wall}}/t_{\text{Pl}}}{(m_T + 2)(m_T + 1)} \approx \frac{t_{\text{wall}}}{t_{\text{Pl}}} \left(\frac{T_{\text{obs}}}{T_{\text{Pl}}} \right)^2.$$

This is the central use of the same stars-and-bars count (1) that delivered the finite-mode Kirchhoff law in §5; we are not introducing a new HQIV input.

Birefringence formula. HQIV’s [cumulative birefringence observable](#), proven zero-sorry, is

$$(15) \quad \beta(m_{\text{emit}}, m_{\text{obs}}) = \alpha \log \left(\frac{m_{\text{obs}} + 1}{m_{\text{emit}} + 1} \right),$$

with $\alpha = 3/5$. Specialising to the near-pole reading ($m_{\text{emit}} = 0$ at the Planck pole, $m_{\text{obs}} = m_{\text{prop}}$):

$$(16) \quad \beta_{\text{HQIV}} = \alpha \log(1 + m_{\text{prop}}).$$

Numerics. Inputs: $T_{\text{Pl}} = 1.41679 \times 10^{32}$ K, $T_{\text{CMB}} = 2.7255$ K (FIRAS [8]), $t_{\text{Pl}} = 5.391 \times 10^{-44}$ s, $t_{\text{wall}} = 51.2$ Gyr (the [wall-clock-age snapshot](#) from the HQIV-modified CLASS background), $\alpha = 3/5$ (informational monogamy). Then

$$\begin{aligned} m_T &= 5.198 \times 10^{31}, \\ \mathcal{N}(m_T) &= 2.702 \times 10^{63}, \\ t_{\text{wall}}/t_{\text{Pl}} &= 2.997 \times 10^{61}, \\ m_{\text{prop}} &= 1.109 \times 10^{-2}. \end{aligned}$$

$$(17) \quad \beta_{\text{HQIV}} = \frac{3}{5} \log(1.01109) = 6.619 \times 10^{-3} \text{ rad} = \boxed{0.3792^\circ},$$

deviation -0.40σ from (10). The exact arithmetic chain reproducing every line of this paragraph ships in the supplementary script bundle (Appendix B, file `birefringence_calculation.py`).

Falsifier: apparent age fails. Substituting the [HQIV apparent age](#) of 13.8 Gyr for t_{wall} : $m_{\text{prop}}^{\text{app}} \approx 3.0 \times 10^{-3}$, $\beta^{\text{app}} \approx 0.103^\circ$, a -2.55σ deviation. The formula *selects* HQIV’s wall-clock age over the apparent age—this is a non-adjustable test of the ADM-lapse subsystem against an independent observation.

Observer-position character (PR4 frequency independence). By (16) and (14), β_{HQIV} depends on $(t_{\text{wall}}, T_{\text{obs}})$ only, not on individual photon frequencies. All photons reaching the same observer get the same imprint, formalised as the [frequency-independence statement for the predicted birefringence](#), and exactly the PR4 finding [1] across 30–353 GHz. A per-photon ω^2 mechanism would predict $\sim 140\times$ amplification across that range and is ruled out; HQIV does not predict that. The blackbody Planck spectrum smears in photon energy; the β value does not.

Isotropy. m_{prop} is a scalar. β_{HQIV} has no preferred direction, matching the PR4 isotropy null result—formalised as the [isotropy lemma](#) for the predicted birefringence (with the [near-pole specialisation](#) used in (16)).

8. FRIEDMANN SCALING RECOVERED, NOT IMPORTED

In a radiation-dominated FLRW cosmology $H \propto T^2/M_{\text{Pl}}$ and so the Hubble time is $t_H = 1/H \propto M_{\text{Pl}}/T^2$. This is famously the quadratic growth law.

In HQIV, identifying one propagation shell with the bundle of new lattice simplices at the observation depth (the same identification that delivered (14)) gives the propagation-shell duration $\sim t_{\text{Pl}} \mathcal{N}(m_T) \sim t_{\text{Pl}}(T_{\text{Pl}}/T)^2 = M_{\text{Pl}}/T^2$ in Planck units.

Theorem 8.1 (Friedmann scaling recovered). *Under the propagation-shell identification of §7,*

$$t_H^{\text{HQIV}}(T) \sim t_{\text{Pl}} \mathcal{N}(m_T) \sim M_{\text{Pl}}/T^2.$$

The same stars-and-bars count (1) that delivers the finite-mode Kirchhoff law also delivers Friedmann’s $H \propto T^2$. No Einstein equation has been used. The arithmetic ratio $t_H^{\text{HQIV}}(T)/t_H^{\text{rad}}(T) = (m_T + 2)/(m_T + 1) \rightarrow 1 + O(1/m_T)$ at present-epoch temperatures is exhibited by the supplementary script `friedmann_recovery.py` described in Appendix B.

9. DISCUSSION: WHERE NORMALISATION WENT

In the standard derivation of the Planck spectrum, a UV regulariser is implicit (a cavity cutoff, dimensional reg, or simply trust in the Bose factor); an IR regulariser is also implicit (the cavity volume). Renormalisation theory then constructs observables independent of these regularisers. This is conceptually expensive—the cutoffs are tools, not physics.

HQIV trades this for two combinatorial inputs. The discrete null lattice provides a structural UV cutoff at $m = 0$ (Planck pole); the lock-in reference shell provides a structural IR cutoff at $m = M$. Both cutoffs are physical choices, not regulators: moving them changes observables in a calculable way, captured by the monotonicity statements of Theorem 5.4. The total mode budget at any pair of cutoffs is the closed-form hockey-stick sum (2).

What is gained:

- No UV catastrophe—the high-frequency modes do not exist.
- No IR divergence—the low-frequency modes are finite in count.
- No renormalisation procedure—the sums are finite as written.
- Under the proton lock-in scale-witness convention, one non-trivial quantitative prediction ($\beta = 0.379^\circ$ at present epoch) survives the data at -0.40σ and discriminates wall-clock from apparent age internally.
- Friedmann scaling recovered as a consequence (Theorem 8.1), not imported.

What is *not* claimed:

- HQIV is not claimed to be independent of every scale anchor. The proton mass at $m = 4$ uses the theory’s internal constituent-plus-binding readout, with the observed value 938.272 MeV serving as the active scale witness. Substituting an alternative witness (CODATA α , present-CMB) would re-pin every dimensionful HQIV output; no such cross-witness consistency demonstration is given here.
- The shell index $m = 4$ is not a two-axiom output. It is the QCD-onset identification, and the constituent-plus-binding closure at that shell to 938.272 MeV is a non-trivial HQIV calculation, but the placement of $m = 4$ itself is a modelling choice.
- The wall-clock age 51.2 Gyr is an output of the HQIV-modified CLASS implementation (a discrete null-lattice background with the ADM lapse and φ -dependent evolution), not a free parameter, but neither is it strictly forced by the two axioms alone—the HQIV-modified CLASS background carries its own modelling choices on top of the lattice.
- One further piece of the β chain is an explicit identification, not a strict consequence of the two axioms: “one propagation shell = one bundle of new lattice simplices at the observation depth” (the same construction that gives Friedmann’s T^2 scaling).

- No continuum-limit programme is invoked, required, or desired anywhere in this chain. Per the patch-theory reader contract, the discrete patch layer is the ontology; the only continuum object that appears is the de-facto continuum algebra $\mathfrak{so}(8)$ obtained from finite octonionic data [5, 6], and even that is built from discrete ingredients with no continuum spacetime attached.

The proper reading of this paper is therefore: HQIV’s combinatorial mode-count axiom delivers Kirchhoff’s law on a finite shell list without any regularisation; under the standing proton lock-in scale-witness convention, combining the HQIV wall-clock age from the HQIV-modified CLASS implementation with the lattice mode count $\mathcal{N}(m)$ under the propagation-shell identification predicts $\beta_{\text{HQIV}} = 0.379^\circ$, surviving the PR4 data [1, 12] to -0.40σ . Stronger claims of full independence require either a derivation of $m = 4$ and t_{wall} from the two axioms or a cross-witness consistency demonstration; both are open theoretical targets.

APPENDIX A. LEAN MODULE INDEX

The Lean development lives in the public slimmed mirror <https://github.com/HQIV/hqiv-lean> [7]. All theorems cited in the body are sorry-free at the time of writing. Each entry below carries a reader-friendly anchor (used by hyperlinks in the body), a one-line description, and clickable links to the Lean module and identifiers.

Lattice simplex mode count.:

The integer-solution count $\mathcal{N}(m) = (m + 2)(m + 1)$ at temperature-ladder depth m . Lean: `latticeSimplexCount` and `latticeSimplexCount_eq` in [Hqiv/Geometry/OctonionicLightCone.lean](#); positivity is recorded as `latticeSimplexCount_pos`.

Hockey-stick cumulative-count identity.:

The closed-form sum $3 \sum_{m'=0}^n \mathcal{N}(m') = (n + 1)(n + 2)(n + 3)$. Lean: `cumLatticeSimplexCount_hockey_stick` and `cumLatticeSimplexCount_closed` in [Hqiv/Geometry/OctonionicLightCone.lean](#).

Auxiliary-field temperature ladder.:

$T(m) = T_{\text{Pl}}/(m + 1)$ on the discrete shell index m . Lean: `T` in [Hqiv/Geometry/AuxiliaryField.lean](#), with `shell0omega` (the harmonic ladder of (3)) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Per-shell Bose factor.:

The standard occupation $n_B(\omega_m, T)$ of (4). Lean: `nBose` (positivity `nBose_pos`) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Per-shell spectral-energy density.:

Equation (5), $E_m(T) = \mathcal{N}(m)\omega_m n_B(\omega_m, T)$. Lean: `shellSpectralEnergy` (and `shellModeMultiplicity` for $\mathcal{N}(m)$ in the new-modes basis) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Truncated blackbody energy density.:

Equation (6), with the three-clause finiteness / positivity / monotonicity package. Lean: `blackbodyEnergyDensity`, `blackbodyEnergyDensity_pos_of_le`, `blackbodyEnergyDensity_mono_UV`, `blackbodyEnergyDensity_mono_IR`, `planckUVCutoff` (Planck pole) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Lock-in IR cutoff.:

The structural infrared cutoff at the lock-in reference shell M , packaged as `referenceIRCutoff` in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Vacuum zero-point cell.:

Equation (7); Lean: `vacuumZeroPointEnergy` (non-negativity `vacuumZeroPointEnergy_nonneg`) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Blackbody-plus-vacuum energy density.:

Equation (8); Lean: `blackbodyEnergyDensityWithVacuum` (and the additive decomposition lemma) in [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Horizon Stefan–Boltzmann module.:

Recovery of the T^4 scaling and Stefan-type ceiling on a finite shell list. Lean: [Hqiv/Physics/HorizonBlackbodyStefan.lean](#).

Horizon Wien-displacement module.:

Wien displacement bound from the same finite-mode spectrum. Lean: [Hqiv/Physics/HorizonBlackbodyWienDisplacement.lean](#) (with the polarized-greybody companion in [Hqiv/Physics/HorizonBlackbodyGreybody.lean](#)).

Proton lock-in scale-witness.:

The HQIV scale-witness convention pinning the mass/unit chart at $m = 4$ to $M_p = 938.272$ MeV. Lean: `ScaleWitness.proton_lockin` (and the `scale-witness` enum) in [Hqiv/Physics/ScaleWitness.lean](#).

Scale-witness enum.:

The Lean enum recording the candidate single-witness anchors (`proton_lockin`, `codata_alpha`, `cmb_now`). Lean: `ScaleWitness` in [Hqiv/Physics/ScaleWitness.lean](#).

Derived proton mass.:

The constituent-minus-binding readout $M_p^{\text{HQIV}} = M_p^{\text{const}} - E_{\text{bind}}^{\text{shared}}$ of (9). Lean: `derivedProtonMass` (positivity `derivedProtonMass_pos`) in [Hqiv/Physics/DerivedNucleonMass.lean](#).

Shared-harmonics proton-mass closure.:

The closure lemma certifying that the constituent and shared pieces sum to 938.272 MeV. Lean: `proton_mass_from_shared_harmonics` in [Hqiv/Physics/DerivedNucleonMass.lean](#).

Proton-anchor mass constant.:

The 938.272 MeV anchor numeral that pins the unit chart. Lean: `protonAnchorMass_MeV` (also `protonMassFromMetaHarmonics_MeV`) in [Hqiv/Physics/QuarkMetaResonance.lean](#).

Cosmological shell-ladder module.:

The propagation-shell coordinate m_{prop} of (14), the simplex-count cast `latticeSimplexCount_as_shell_coarseness`, and the internal-coordinate identity `m_prop_HQIV_internal_eq`. Lean: [Hqiv/Cosmology/CosmologicalShellLadder.lean](#).

Cumulative birefringence observable.:

Equation (15); the near-pole specialisation (16) is `NearPoleObservation.predictedBirefringence`, and the numerical pass against PR4 errors is `nearPole_cmb_shell_ladder_pass`. Lean: [Hqiv/Cosmology/CosmologicalShellLadder.lean](#) and [Hqiv/Physics/HorizonBlackbodyLadder.lean](#).

Isotropy of the predicted birefringence.:

Lean: `predictedBirefringence_isotropic` (with the `near-pole specialisation`) in [Hqiv/Cosmology/CosmologicalShellLadder.lean](#); the no-shell-crossing zero is `predictedBirefringence_zero_of_no_shell_crossing`, and the lab-scale upper bound is `nearPole_predictedBirefringence_lab_bound`.

Near-pole isotropy specialisation.:

Lean: `nearPole_predictedBirefringence_isotropic` in [Hqiv/Cosmology/CosmologicalShellLadder.lean](#).

Frequency-independence of the predicted birefringence.:

All photons reaching the same observer carry the same birefringence imprint. Lean: `predictedBirefringence_frequency_independent` in [Hqiv/Cosmology/CosmologicalShellLadder.lean](#).

HQVM wall-clock compression.:

The HQIV-modified CLASS background output recording $t_{\text{wall}} = 51.2$ Gyr for downstream proof reuse. Lean: `age_wall_clock_Gyr_paper` (with `wallClockAgeHomogeneous` and `HQVM_lapse`) in [Hqiv/Geometry/UniverseAge.lean](#).

HQIV apparent age.:

The companion snapshot 13.8 Gyr used as the wall-clock falsifier in §7. Lean: `age_apparent_Gyr_paper` in [Hqiv/Geometry/UniverseAge.lean](#).

APPENDIX B. REPRODUCER SCRIPT BUNDLE

The numerical results in this paper—the $\beta_{\text{HQIV}} = 0.379^\circ$ prediction and its -0.40σ agreement with Planck PR4 [1, 12]; the 0.103° apparent-age falsifier; the finite-mode Kirchhoff energy density at FIRAS T_{CMB} [8]; and the Friedmann T^2 recovery (Theorem 8.1)—are reproduced by a small, dependency-light script bundle shipped alongside the manuscript as `scripts.zip`. The bundle is intended for upload as a Zenodo supplementary file together with the paper PDF so that it shares the manuscript DOI and is permanently citable.

Contents.

birefringence_calculation.py: Reproduces every line of the §7 numerics paragraph: $m_T, \mathcal{N}(m_T), t_{\text{wall}}/t_{\text{Pl}}, m_{\text{prop}}, \beta_{\text{HQIV}}$ in radians and degrees, residual to PR4 in σ units, plus the apparent-age falsifier ($\beta^{\text{app}} = 0.103^\circ, -2.55\sigma$).

kirchhoff_finite_mode.py: Computes the truncated blackbody energy density of (6) on the FIRAS CMB temperature for a small UV/IR window, demonstrating the monotonicity statements of Theorem 5.4 and the hockey-stick cumulative count of (2) numerically.

friedmann_recovery.py: Confirms the Theorem 8.1 scaling $t_H^{\text{HQIV}} \sim t_{\text{Pl}} \mathcal{N}(m_T) \sim M_{\text{Pl}}/T^2$ for several reference temperatures (CMB today, recombination, big-bang nucleosynthesis), and reports the relative discrepancy with the standard radiation-era Friedmann formula at each temperature.

README.md: How-to-run instructions, contents table, patch-ontology disclaimer (the scripts are interoperability tools, not a continuum-limit refinement), and a citation block.

Requirements. Python 3.10+ with the standard library only. No third-party packages, no random seeds, and no external data files; the bundle is fully deterministic and runs in well under a second on any modern laptop.

Patch-ontology disclaimer. The numerical outputs are calculation-approximation translations of patch quantities (cycle indices, shell ladders, normalised readouts) into degrees, kelvin, and SI energies. They are *not* a continuum-limit refinement of the patch theory; they are the literature-translation rewrites called for by the patch-theory reader contract above, packaged so that any reader can re-execute them on their own machine.

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¹AI-assisted drafting and repository tooling were used during manuscript preparation; all formal claims refer to the HQIV-LEAN library, and the author reviewed hypothesis lists in the cited Lean modules.

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